Hands-On-Activity No. 6.2	
Title **Files and Streams**	
Course Code: CPE 007	**Program: Computer Engineering**
Course Title: Programming Logic and Design	**Date Performed: Oct 27, 2026**
Section: 11S2	**Date Submitted: Oct 27, 2026**
Name: Torralba, Blizhel	**Instructor: Noel Jason Lusung**

Objective(s)

This activity aims to develop programming skills using the concepts of files and streams.

Intended Learning Outcomes

The students should be able to:
1. **Work with files**
2. **Read from a file**
3. **Write to a file**
4. **Process simple data**
5. **Understand the concept of the loop over file content**

Materials and Equipment

Internet connection
Personal Computer

Procedure

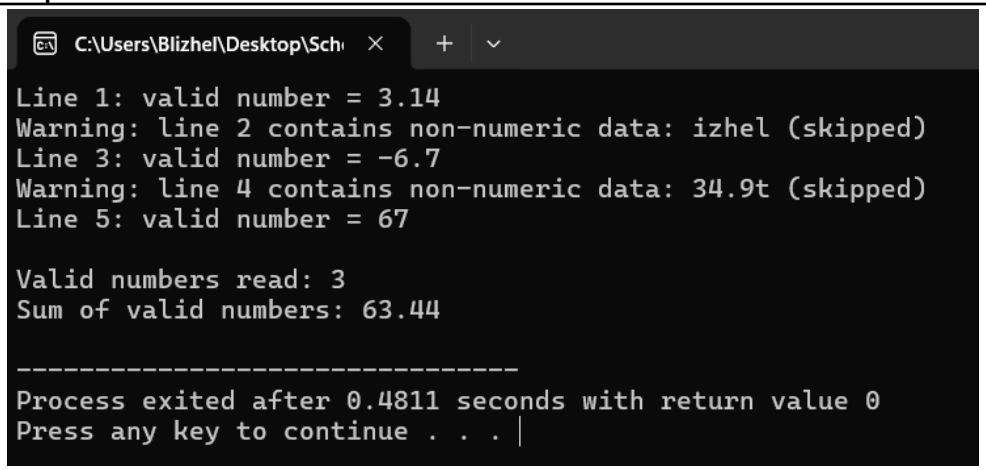
Directions: Use Dev-C++ for the program and include comments for explanations.

1. **Create a program that reads floating-point numbers line-by-line from a text file. If non-numeric characters are encountered on a line, log a warning, clear the error, and skip to the next line without crashing.**
2. **Write a program that opens a text file containing lines of mixed numbers and words. Read through the file line by line, print valid numbers, and handle/skip lines containing invalid text data without halting execution.**

Submit a document with the format CPE007_Seatwork6_FullName.docx along with the program in two (2) .cpp file formats.

Output

1.




```
#include <iostream>
#include <fstream>
#include <sstream>
#include <string>
```

```
using namespace std;

int main()
{
   ifstream inFile("numbers.txt"); // opens file

   if (!inFile) { // stops if file is missing
      cerr << "Error: could not open numbers.txt!" << endl;
      return 1;
   }

   string line;
   int lineNumber = 0;
   int validCount = 0;
   double sum = 0.0;

   while (getline(inFile, line)) { // read one line at a time
      lineNumber++;

      if (line.find_first_not_of(" \t\r") == string::npos) {
         continue;
      }

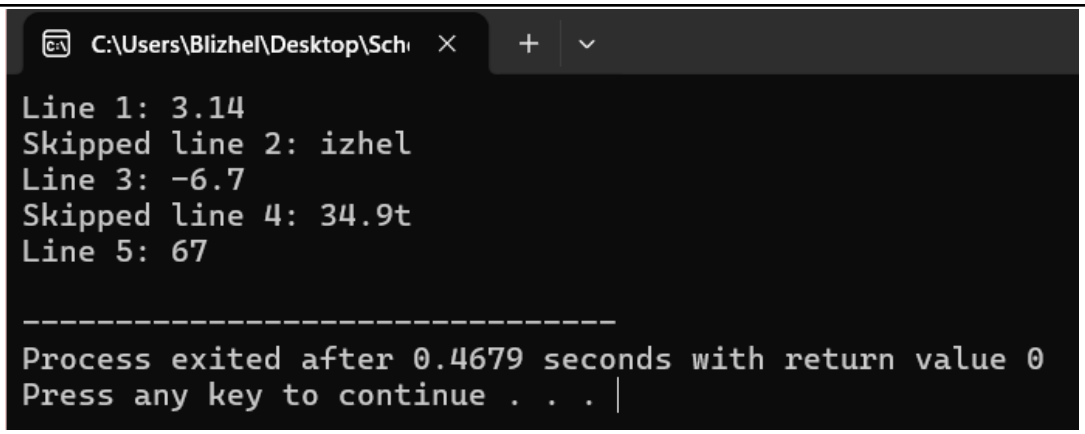
      istringstream iss(line);
      double value;
      string leftover;

      if ((iss >> value) && !(iss >> leftover)) { // valid only if a number is read
         cout << "Line " << lineNumber << ": valid number = " << value << endl;
         sum += value;
         validCount++;
      } else { // warns and go to next line
         cerr << "Warning: line " << lineNumber
              << " contains non-numeric data: " << line << " (skipped)" << endl;
         iss.clear();
      }
   }

   inFile.close(); // close file

   cout << "\nValid numbers read: " << validCount << endl;
   cout << "Sum of valid numbers: " << sum << endl;

   return 0;
}
```



2.

```
#include <iostream>
#include <fstream>
#include <sstream>
#include <string>
using namespace std;

int main()
{
   ifstream inFile("numbers.txt"); // open input file
   if (!inFile) {
      cerr << "Error: could not open numbers.txt!" << endl;
      return 1;
   }

   string line;
   int lineNumber = 0;

   while (getline(inFile, line)) { // read line by line
      lineNumber++;

      // skip blank lines
      if (line.find_first_not_of(" \t\r") == string::npos) {
         continue;
      }

      istringstream iss(line);
      double value;
      string leftover;

      // valid only if a number is read and nothing follows it
      if ((iss >> value) && !(iss >> leftover)) {
         cout << "Line " << lineNumber << ": " << value << endl;
      } else {
         cerr << "Skipped line " << lineNumber << ": " << line << endl;
      }
   }

   return 0;
```

```
}
```

Conclusion

In this activity I learned how to create a program with the concepts of files and streams.